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Engine nacelle inlet having a turbulator segment

Abstract

A nacelle inlet for an engine. The nacelle inlet includes an inner flow surface, an outer flow surface, and a turbulator segment positioned along the inner flow surface. The turbulator segment extends outward beyond the inner flow surface and is configured to transition laminar airflow along the inner flow surface upstream from the turbulator segment to turbulent airflow downstream from the turbulator segment.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3578264	12/1970	Kueth	181/220	F28F 1/32
3664612	12/1971	Skidmore et al.	137/15.1	B64D 33/02
3819008	12/1973	Evans	428/116	F02C 7/045
4154256	12/1978	Miller	137/15.1	B64D 33/02
5156362	12/1991	Leon	244/130	B64D 33/02
6655632	12/2002	Gupta	244/1N	B64C 23/06
7878759	12/2010	Mills	416/61	F04D 29/668
8408491	12/2012	Jain	415/126	F02C 7/042
8529188	12/2012	Winter	415/126	F02C 7/042
9273558	12/2015	Liang	N/A	N/A
9725190	12/2016	Batt et al.	N/A	N/A
10294867	12/2018	Lumbab et al.	N/A	N/A
11702982	12/2022	Gelzer et al.	N/A	N/A
2005/0274103	12/2004	Prasad	137/15.1	B64D 33/02
2009/0155067	12/2008	Haas	415/208.1	B64D 33/02
2021/0231054	12/2020	Dindar	N/A	F02C 7/057
2022/0099023	12/2021	Pascal	N/A	B64D 15/04
2022/0356839	12/2021	LaBelle	N/A	F01D 25/243

OTHER PUBLICATIONS

Davis, W., et al., "Infrared thermography techniques for boundary layer state visualisation", XXVI Biennial Symposium on Measuring Techniques in Turbomachinery, Journal of Physics: Conference Series, Aug. 24, 2023, pp. 1-16. cited by applicant

Luckring, J., et al., "An Application of CFD to Guide Forced Boundary-Layer Transition for Low-Speed Tests of a Hybrid Wing-Body Configuration", AIAA 34th Applied Aerodynamics Conference, Jan. 1, 2016, pp. 1-23, Washington, DC. cited by applicant

Van Nesselrooij, M., et al., "Drag reduction in turbulent boundary layers by means of dimpled surfaces", Delft University of Technology, Faculty of Aerospace Engineering, Dec. 4, 2017, pp. 1-19. cited by applicant

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Background/Summary

TECHNOLOGICAL FIELD

(1) The present disclosure relates generally to the field of engine nacelles and, more specifically, the present disclosure relates to the field of transitioning airflow from laminar to turbulent within an engine nacelle inlet to facilitate cooling.

BACKGROUND

- (2) An engine nacelle is a housing that extends around an engine on an aircraft. The nacelle includes an inlet that directs the air to the engine. The nacelle inlet functions to protect the engine, such as a gas turbine engine, from ingesting foreign objects. The nacelle inlet is further configured to direct air to the engine. The nacelle inlet is heated during aircraft use to prevent icing. One method for inlet heating occurs by directing heated air from the engine into one or more interior sections of the nacelle inlet. An issue with the heating is that one or more sections of the nacelle inlet can reach an excessively elevated temperature. One location of the nacelle inlet that can experience excessively elevated temperatures is along an inner flow surface of the nacelle inlet.
- (3) Cooling of the one or more sections of the nacelle inlet occurs as a result of heat transfer to the air that enters the engine through the nacelle inlet. As a product of the air flowing along the surfaces, heat is drawn out of the one or more sections which reduces the surface temperature. The inner flow surface of the nacelle inlet is configured to promote laminar airflow over the one or more sections. However the boundary layer is relatively slow moving and retains the heat drawn from the heated lipskin and carries this heat downstream. Turbulent airflow disrupts this boundary layer heat transfer effect by promoting mixing with the adjacent cooler airflow, thus reducing the heat transfer into adjacent structure downstream.
- (4) Thus, there is a need to transition airflow from laminar to turbulent locally within the nacelle inlet to reduce excessive temperature of adjacent structure downstream of the heated sections.

SUMMARY

- (5) One aspect is directed to a nacelle inlet for an engine. The nacelle inlet comprises an inner flow surface, an outer flow surface, and a turbulator segment positioned along the inner flow surface. The turbulator segment extends outward beyond the inner flow surface and is configured to transition laminar airflow along the inner flow surface upstream from the turbulator segment to turbulent airflow downstream from the turbulator segment.
- (6) In another aspect, the turbulator segment is a strip that is connected to the inner flow surface with the strip comprising an inner face positioned towards the inner flow surface, an opposing outer face, a leading edge, and a trailing edge.
- (7) In another aspect, the strip extends completely around a circumference of the nacelle inlet.
- (8) In another aspect, the strip comprises a plurality of discrete sections that are spaced apart by gaps.
- (9) In another aspect, the turbulator segment comprises a plurality of fasteners with the plurality of fasteners comprising heads that extend outward beyond the inner flow surface.
- (10) In another aspect, the heads are aligned in a row around a circumference of the nacelle inlet.
- (11) In another aspect, the turbulator segment is positioned on one or more members that are connected to the inner flow surface.
- (12) In another aspect, the one or more members are constructed from materials that are different from the inner flow surface.
- (13) In another aspect, the turbulator segment comprises one or more strips and fasteners that extend through the one or more strips and into the inner flow surface.
- (14) One aspect is directed to a nacelle inlet for an engine. The nacelle inlet comprises a lipskin positioned at a forward end of the nacelle inlet with the lipskin comprising a lipskin inner flow surface that extends to an inner lipskin edge. An inner barrel is positioned downstream from the lipskin inner flow surface. A joint is formed between the inner lipskin edge and a leading edge of the inner barrel. A turbulator segment is positioned at the lipskin inner flow surface upstream from the joint with the turbulator segment configured to transition laminar airflow to turbulent airflow downstream from the joint to draw heat from the inner barrel.
- (15) In another aspect, the inner barrel comprises inner barrel structural plies that are constructed from composite material.
- (16) In another aspect, the turbulator segment is a strip that is connected to the nacelle inlet and that extends outward beyond the lipskin inner flow surface.

- (17) In another aspect, the strip is a continuous band that extends completely around a circumference of the nacelle inlet.
- (18) In another aspect, the turbulator segment comprises a leading edge and a trailing edge with the trailing edge aligned with the joint.
- (19) In another aspect, the turbulator segment comprises a leading edge and a trailing edge and with the turbulator segment positioned forward of the joint such that the trailing edge is spaced away from the inner lipskin edge by a distance.
- (20) In another aspect, a bulkhead spans a lipskin annulus with the bulkhead comprising a flange that extends across an inner side of the joint.
- (21) In another aspect, the turbulator segment comprises a plurality of fastener heads that extend outward beyond the inner flow surface.
- (22) In another aspect, the lipskin inner flow surface is substantially smooth.
- (23) In another aspect, the turbulator segment is a trough in the lipskin inner flow surface.
- (24) One aspect is directed to a method of cooling a section of an inner flow surface of a nacelle inlet. The method comprises: directing air through the nacelle inlet with laminar flow along an upstream section of the inner flow surface; directing the air over a raised turbulator segment on the inner flow surface after the air has passed over the upstream section and transitioning the air to turbulent flow; and directing the air with turbulent flow over a downstream section of the inner flow surface of the nacelle inlet.
- (25) In another aspect, the method further comprises directing the air over a strip that is attached to the inner flow surface and causing laminar flow to transition to turbulent flow.
- (26) The features, functions and advantages that have been discussed can be achieved independently in various aspects or may be combined in yet other aspects, further details of which can be seen with reference to the following description and the drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an isometric view of an aircraft with nacelles positioned underneath each of the wings.
- (2) FIG. 2 is a perspective view of a nacelle inlet.
- (3) FIG. 3 is a schematic section view of a portion of the nacelle having an inner flow surface leading into the engine and an opposing outer flow surface.
- (4) FIG. 4 is an enlarged view of the nacelle inlet portion of FIG. 3.
- (5) FIG. 5 is a schematic section view of a strip attached to a lipskin and forming a turbulator segment upstream from an inner barrel edge band.
- (6) FIG. 6 is a side schematic view of a strip that forms a turbulator segment.
- (7) FIG. 7 is a schematic section view of a turbulator segment positioned on an inner flow surface of a nacelle inlet.
- (8) FIG. 8 is a schematic section view of a fastener attached to a lipskin and forming a turbulator segment upstream from an inner barrel edge band.
- (9) FIG. 9 is a schematic section view of a turbulator segment positioned on an inner flow surface of a nacelle inlet.
- (10) FIG. 10 is a schematic section view of an enlarged section of a lipskin forming a turbulator segment upstream from an inner barrel edge band.
- (11) FIG. 11 is a schematic section view of a trough in a lipskin and forming a turbulator segment upstream from an inner barrel edge band.
- (12) FIG. 12 is a flowchart diagram of a method of cooling a section of an inner flow surface of a nacelle inlet.

DETAILED DESCRIPTION

(13) FIG. 1 illustrates an aircraft **100** configured to transport people and/or cargo. The aircraft **100** generally includes a fuselage **101** and wings **102**. Engines **103** are mounted on the wings **102** to propel the aircraft **100** during flight. The number and positioning of the engines **103** can vary depending upon the aircraft **100**. In some examples, the engines **103** are gas turbine engines, such as turbofan engines.

(14) The engines **103** include a nacelle **20** that extends around the exterior. The nacelle **20** has an aerodynamic profile such as being round or elliptical to reduce drag. The nacelle **20** includes a nacelle inlet **21** that directs airflow into the engine **103** and with a centerline C/L that extends through the nacelle inlet **21**. As illustrated in FIGS. 2 and 3, the nacelle inlet **21** includes a forward end **60**, an inner flow surface **61** that extends along the inlet **21**, and an outer flow surface **62** that extends along the exterior side. The inner flow surface **61** and outer flow surface **62** can be formed by one or more different components.

(15) The forward section of the nacelle inlet **21** including the forward end **60** is formed by a lipskin **22**. The lipskin **22** forms portions of both the inner flow surface **61** and the outer flow surface **62**. Specifically the lipskin **22** includes a lipskin inner flow surface **23** that extends between an inner lipskin edge **24** and hilite **25**. The lipskin **22** further includes a lipskin outer flow surface **26** that extends from the hilite **25** to an outer lipskin edge **27**.

(16) An inner barrel **30** extends aft from the inner lipskin edge **24**. The inner barrel **30** includes inner barrel structural plies **32**, a composite core **33**, and an inner barrel perforated skin **36**. The inner barrel structural plies **32** include an inner barrel edge band **34** that extends between a leading edge **31** and an inner barrel ramp transition **35**. The inner barrel edge band **34** forms a portion of the inner flow surface **61**. The leading edge **31** of the inner barrel **30** forms a joint **91** with the inner lipskin edge **24**. A composite core **33** is bonded to the inner barrel structural plies **32** aft of the inner barrel ramp transition **35**. The inner barrel **30** can include various other constructions and configurations than those illustrated herein.

(17) A generally annular forward bulkhead **40** extends radially across a lipskin annulus **47**. In some examples, the forward bulkhead **40** includes a flange **41** that extends across the joint **91**. In other examples, the flange **41** is a separate member from the forward bulkhead **40**. A duct **45** is formed between the bulkhead **40** and the lipskin **22** at the forward section of the nacelle inlet **21**. In addition, a generally annular aft bulkhead (not illustrated) extends radially between the inner barrel **30** and the outer barrel **28**.

(18) The nacelle inlet **21** is configured to direct airflow to the engine **103**. Airflow that enters through the nacelle inlet **21** flows in the direction of arrow A along the inner flow surface **61** that is formed by the lipskin inner flow surface **23**, inner barrel edge band **34**, and the inner barrel perforated skin **36**.

(19) The lipskin inner flow surface **23** is heated to prevent icing on the surface. In some examples, the heat is provided from high temperature bleed air that is routed from the engine **103** and directed into the duct **45** formed at the forward section of the nacelle inlet **21**. In some examples, the bleed air has a temperature of about 1000° F. The heated air directed to this area can result in one or more of the elements in the nacelle inlet **21** being heated to an elevated temperature. In some examples, the forward bulkhead **40** is heated to a temperature of about 740° F. and the flange **41** is heated to a temperature range of 350° F.-500° F. Furthermore, the temperature of the inner barrel edge band **34** is elevated due to conduction with the flange **41**. In some examples, the temperature of the inner barrel edge band **34** may approach a range of approximately 350° F.-500° F.

(20) Airflow across the inner flow surface **61** removes heat from the inner barrel edge band **34** and the lipskin **22**. The surface of the lipskin **22** is substantially smooth which results in laminar airflow across the inner flow surface **61**. Heat transfer occurs between the lipskin **22** and laminar airflow which raises the temperature of the airflow boundary layer along the inner flow surface **61**. This heat is carried downstream to the inner barrel edge band **34** raising the temperature of the inner barrel edge band **34**. To address the issue of heat removal, a turbulator segment **50** at the lipskin

inner flow surface **23** transitions laminar airflow to turbulent airflow. Turbulent airflow is directed over one or more of the sections of the downstream inner flow surface **61** to disrupt the boundary layer heat transfer effect from the one or more components. Turbulent airflow draws the heat from the one or more components and is more effective at reducing temperature than laminar airflow due to boundary layer mixing. In one example, turbulent airflow reduces the temperature of the inner barrel edge band **34**.

(21) In some examples, the turbulator segment **50** is formed on the nacelle inlet **21** during manufacturing. In other examples, the process is performed on existing aircraft that are in use. The process utilizes tools and machinery that facilitate retrofitting existing aircraft.

(22) The turbulator segment **50** can include a variety of different configurations. The turbulator segment **50** can be positioned at various distances relative to the inner lipskin edge **24**. In some examples, the turbulator segment **50** is spaced a distance d away from the inner lipskin edge **24**. In other examples, the turbulator segment **50** is positioned at the inner lipskin edge **24**.

(23) In one example as illustrated in FIG. 5, the turbulator segment **50** is a strip **70** that is attached to the lipskin inner flow surface **23**. The strip **70** includes an inner face **74** that contacts against the lipskin inner flow surface **23** and an opposing outer face **75** that is exposed. The strip **70** also includes a leading edge **72** and a trailing edge **73**. The strip **70** includes a thickness measured between the inner face **74** and the outer face **75** that is sized to extend outward beyond the surface of the lipskin inner flow surface **23**. The thickness can be constant throughout the strip **70** as illustrated in FIG. 5 or can vary along one or more sections.

(24) One or more of the leading edge **72** and the trailing edge **73** can be configured to facilitate turbulent airflow. As illustrated in FIG. 6, the leading edge **72** is aligned at an angle α with the inner face **74** and the trailing edge **73** is aligned at an angle β with the inner face **75**. In some examples as illustrated in FIG. 6, the angles α , β are acute. In some examples as illustrated in FIG. 5, the angles α , β are at substantially 90. The angles α , β can be the same as illustrated in FIGS. 5 and 6 or can be different. In some examples, one or both of the leading edge **72** and trailing edge **73** are flat. In other examples, one or both include a variable surface with examples including but not limited to one or more curves and/or are rounded. The strip **70** includes a width W measured between the leading edge **72** and the trailing edge **73**. The width W can be constant along the nacelle inlet **21** or can vary.

(25) The strip **70** can be positioned at various locations along the lipskin inner flow surface **23**. In some examples as illustrated in FIG. 5, the strip **70** is aligned at the joint **91** with the trailing edge **73** of the strip **70** aligned with the inner lipskin edge **24**. In other examples, the strip **70** is positioned at various spacings that are forward of the inner lipskin edge **24**.

(26) In some examples, the strip **70** is a continuous band that extends completely around the nacelle inlet **21**. In other examples, the strip **70** is formed by one or more discrete sections **76**. FIG. 7 illustrates an example with the strip **70** formed by sections **76a**, **76b**, **76c**. The sections **76** are spaced apart by gaps. In some examples, the sections **76** are positioned completely around the circumference of the nacelle inlet **21**. In other examples, the sections **76** are positioned along one or more discrete sections of the nacelle inlet **21**. The sections **76** can have the same or different shapes, sizes, and constructions.

(27) In some examples, the one or more sections **76** are formed by fasteners **77** that extend outward beyond the lipskin inner flow surface **23**. FIG. 8 illustrates an example with a fastener **77** connected to the lipskin **22**. The fastener **77** includes a head **78** that is positioned outward beyond the lipskin inner flow surface **23**. The fastener **77** also includes a body **79** that extends into and/or through one or more of the lipskin **22** and flange **41**. In some examples, a number of fasteners **77** are aligned in a row that extends around a portion or entirety of the circumference of the nacelle inlet **21**.

(28) In some examples, the turbulator segment **50** includes a single strip **70**. In other examples, the turbulator segment **50** includes two or more strips **70**. FIG. 9 illustrates an example with a pair of

strips **70a**, **70b** that are each connected to the lipskin **22**. In one example as shown in FIG. **9**, each of the strips **70a**, **70b** is continuous around the circumference of the nacelle inlet **21**. In other examples, the strips **70a**, **70b** include different shapes, sizes, and/or constructions. FIG. **9** is an example of a turbulator segment **50** with a pair of strips **70a**, **70b**. Other examples include more than three separate strips **70**.

(29) The one or more strips **70** and sections **76** can be constructed from a variety of different materials, including but not limited to aluminum, composite materials, and plastics. The strips **70** are connected to the nacelle inlet **21** in various manners such as but not limited to one or more of mechanical fasteners and adhesives.

(30) FIG. **10** illustrates an example in which the turbulator segment **50** is formed by an enlarged section **80** of the lipskin **22**. In some examples, the enlarged section **80** has a greater thickness than a remainder of the lipskin **22**. This thickness results in the enlarged section **80** extending outward a greater distance than the remainder of the lipskin inner flow surface **23** and the edge band **34**. The enlarged section **80** can extend around a limited section or entirety of the circumference of the nacelle inlet **21**. In some examples as illustrated in FIG. **10**, the enlarged section **80** is positioned at the inner lipskin edge **24**. In other examples, the enlarged section **80** is spaced forward from the inner lipskin edge **24**.

(31) In some examples disclosed above, the turbulator segment **50** includes one or more members that extend outward from the lipskin inner flow surface **23** to cause turbulent airflow. In other examples, the turbulator segment **50** includes one or more troughs **81** in the lipskin **22**. FIG. **11** illustrates an example with a trough **81** formed in the lipskin **22**. The trough **81** has a width measured between the leading edge **82** and the trailing edge **83**. The depth of the trough **81** below the lipskin inner flow surface **23** can vary. In one example, each of the leading edge **82** and the trailing edge **83** are flat. In other examples, one or both of the leading edge **82** and trailing edge **83** include curved, angular, or other non-flat shapes.

(32) In some examples, the turbulator segment **50** includes a single design feature (e.g., just one or more strips **70**, just one or more sections **76**, just one or more troughs **81**). In other examples, the turbulator segment **50** includes two or more different design features. For example, the turbulator segment **50** includes a combination of strips **70**, and/or sections **76**, and/or troughs **81**. In one example, the turbulator segment **50** includes a first design feature (e.g., strip **70**) along a first section of the nacelle inlet **21** and a second design feature (e.g., sections **76**) along a second section of the nacelle inlet **21**.

(33) The turbulator segment **50** creates a surface discontinuity along the lipskin inner flow surface **23**. The turbulator segment **50** is positioned along the lipskin inner flow surface **23** to induce turbulent airflow to one or more downstream sections of the inner flow surface **61**. Turbulent airflow promotes cooling of one or more downstream sections of the nacelle inlet **21**, including the lipskin **22**, inner barrel edge band **34**, inner barrel perforated skin **36**, inner barrel structural plies **32**, and composite core **33**. This cooling occurs due to mixing the boundary layer with the cooler adjacent airflow as a result of the transition from laminar to turbulent. Cooling can also occur to one or more of the components through conduction. In one example, cooling of one or both of the lipskin inner flow surface **23** and the inner barrel edge band **34** results in cooling of the bulkhead flange **41**. In some examples, the lipskin inner flow surface **23** upstream and downstream from the turbulator segment **50** is substantially smooth.

(34) In one example, one or more of the components of the inner flow surface **61** are constructed from a composite ply material. In one specific example, the structural plies **32** including the inner barrel edge band **34** is constructed from a composite ply material. The composite ply material includes one or more layers of fibers that are impregnated with one or more of a thermoset and thermoplastic matrix resin. The fibers can consist of a variety of materials, including but not limited to aramids, polyolefins, metal, glass, carbon, boron, ceramic, mineral, and combinations. The fibers are impregnated with a thermoset or thermoplastic matrix resin. In another example, the

matrix resin includes a hybrid system of both thermoset and thermoplastic resin. The matrix resin can consist of a variety of substances, including but not limited to acrylics, fluorocarbons, polyamides (PA), polyethylenes (PE) such as polyethylene terephthalate (PET), polyesters, polypropylenes (PP), polycarbonates (PC), polyurethanes (PU), polyetheretherketones (PEEK), polyetherketoneketones (PEKK), polyetherimides (PEI), and other material compositions. This construction from a composite material prescribes a maximum operating limit (MOL) temperature and a maximum short duration limit temperature. Turbulent airflow induced by the turbulator segment **50** promotes the airflow to cool the inner barrel edge band **34** to remain below these limits.

(35) FIG. **12** illustrates a method of cooling a section of an inner flow surface **61** of a nacelle inlet **21**. The method includes directing air through the nacelle inlet **21** with the air initially having laminar flow along an upstream section of the inner flow surface **61** (block **200**). The air is directed over a raised turbulator segment **50** on the inner flow surface **61** after the air has passed over the upstream section and transitioning the air to turbulent flow (block **202**). The air with turbulent flow is directed over a downstream section of the inner flow surface **61** of the nacelle inlet **21** (block **204**).

(36) In the examples disclosed above, the nacelles **20** house engines **103** and are used on aircraft. It should be understood, however, that the disclosure applies equally to nacelles **20** for other types of engines in other applications, such as but not limited to other vehicles such as but not limited to landcraft, watercraft, and spacecraft, and powerplant applications.

(37) By the term “substantially” with reference to amounts or measurement values, it is meant that the recited characteristic, parameter, or value need not be achieved exactly. Rather, deviations or variations, including, for example, tolerances, measurement error, measurement accuracy limitations, and other factors known to those skilled in the art, may occur in amounts that do not preclude the effect that the characteristic was intended to provide.

(38) The present invention may, of course, be carried out in other ways than those specifically set forth herein without departing from essential characteristics of the invention. The present embodiments are to be considered in all respects as illustrative and not restrictive, and all changes coming within the meaning and equivalency range of the appended claims are intended to be embraced therein.

Claims

1. A nacelle inlet for an engine, the nacelle inlet comprising: an inner flow surface; an outer flow surface; a turbulator segment positioned along the inner flow surface, wherein the turbulator segment comprises a plurality of fasteners, the plurality of fasteners comprising heads that extend outward beyond the inner flow surface; and wherein the turbulator segment extends outward beyond the inner flow surface and is configured to transition laminar airflow along the inner flow surface upstream from the turbulator segment to turbulent airflow downstream from the turbulator segment.
2. The nacelle inlet of claim 1, wherein the turbulator segment is a strip that is connected to the inner flow surface, the strip comprising an inner face positioned towards the inner flow surface, an opposing outer face, a leading edge, and a trailing edge.
3. The nacelle inlet of claim 2, wherein the strip extends completely around a circumference of the nacelle inlet.
4. The nacelle inlet of claim 2, wherein the strip comprises a plurality of discrete sections that are spaced apart by gaps.
5. The nacelle inlet of claim 1, wherein the heads are aligned in a row around a circumference of the nacelle inlet.
6. The nacelle inlet of claim 1, wherein the turbulator segment is formed on one or more members

that are constructed from materials that are different from the inner flow surface.

7. The nacelle inlet of claim 1, wherein the turbulator segment comprises: one or more strips; and fasteners that extend through the one or more strips and into the inner flow surface.

8. A nacelle inlet for an engine, the nacelle inlet comprising: a lipskin positioned at a forward end of the nacelle inlet, the lipskin comprising a lipskin inner flow surface that extends to an inner lipskin edge; an inner barrel positioned downstream from the lipskin inner flow surface; a joint formed between the inner lipskin edge and a leading edge of the inner barrel; and a turbulator segment positioned at the lipskin inner flow surface upstream from the joint, the turbulator segment configured to transition laminar airflow to turbulent airflow downstream from the joint to draw heat from the inner barrel.

9. The nacelle inlet of claim 8, wherein the inner barrel comprises inner barrel structural plies that are constructed from composite material.

10. The nacelle inlet of claim 8, wherein the turbulator segment is a strip that is connected to the nacelle inlet and that extends outward beyond the lipskin inner flow surface.

11. The nacelle inlet of claim 10, wherein the strip is a continuous band that extends completely around a circumference of the nacelle inlet.

12. The nacelle inlet of claim 8, wherein the turbulator segment comprises a leading edge and a trailing edge and with the trailing edge aligned with the joint.

13. The nacelle inlet of claim 8, wherein the turbulator segment comprises a leading edge and a trailing edge with the turbulator segment positioned forward of the joint with the trailing edge spaced away by a distance from the inner lipskin edge.

14. The nacelle inlet of claim 8, further comprising a bulkhead that spans a lipskin annulus, the bulkhead comprising a flange that extends across an inner side of the joint.

15. The nacelle inlet of claim 8, wherein the turbulator segment comprises a plurality of fastener heads that extend outward beyond the lipskin inner flow surface.

16. The nacelle inlet of claim 8, wherein the lipskin inner flow surface is substantially smooth.

17. The nacelle inlet of claim 8, wherein the turbulator segment is a trough in the lipskin inner flow surface.

18. A method of cooling a section of an inner flow surface of a nacelle inlet, the method comprising: directing air through the nacelle inlet with laminar flow along an upstream section of the inner flow surface; directing the air over a raised turbulator segment on the inner flow surface after the air has passed over the upstream section and transitioning the air to turbulent flow, the turbulator surface comprising one or more strips and fasteners that extend through the one or more strips; and directing the air with turbulent flow over a downstream section of the inner flow surface of the nacelle inlet.

19. The method of claim 18, further comprising directing the air over the strip that is attached to the inner flow surface and causing the laminar flow to transition to turbulent flow.

20. The nacelle inlet of claim 8, further comprising a flange that extends across an inner side of the joint.
